

Configuring IP addressing

About IP addressing

The IP addresses in this chapter refer to IPv4 addresses unless otherwise specified.

IP address representation and classes

IP addressing uses a 32-bit address to identify each host on an IPv4 network. To make addresses easier to read, they are written in dotted decimal notation, each address being four octets in length. For example, address 00001010000000001000000010000001 in binary is written as 10.1.1.1.

Each IP address breaks down into the following sections:

- **Net ID**—Identifies a network. The first several bits of a net ID, known as the class field or class bits, identify the class of the IP address.
- **Host ID**—Identifies a host on a network.

IP addresses are divided into five classes, as shown in [Figure 1](#). The shaded areas represent the address class. The first three classes are most commonly used.

Figure 1 IP address classes

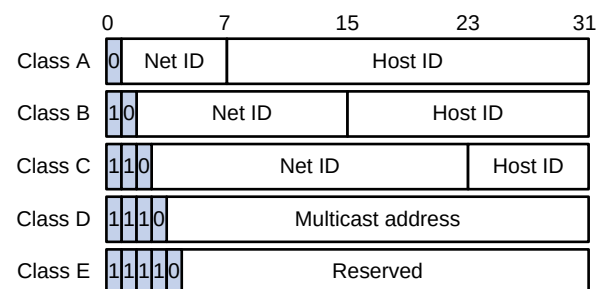


Table 1 IP address classes and ranges

Class	Address range	Remarks
A	0.0.0.0 to 127.255.255.255	The IP address 0.0.0.0 is used by a host at startup for temporary communication. This address is never a valid destination address. Addresses starting with 127 are reserved for loopback test. Packets destined to these addresses are processed locally as input packets rather than sent to the link.
B	128.0.0.0 to 191.255.255.255	N/A
C	192.0.0.0 to 223.255.255.255	N/A
D	224.0.0.0 to 239.255.255.255	Multicast addresses.
E	240.0.0.0 to 255.255.255.255	Reserved for future use, except for the broadcast address 255.255.255.255.

Special IP addresses

The following IP addresses are for special use and cannot be used as host IP addresses:

- **IP address with an all-zero net ID**—Identifies a host on the local network. For example, IP address 0.0.0.16 indicates the host with a host ID of 16 on the local network.
- **IP address with an all-zero host ID**—Identifies a network.
- **IP address with an all-one host ID**—Identifies a directed broadcast address. For example, a packet with the destination address of 192.168.1.255 will be broadcast to all the hosts on the network 192.168.1.0.

Subnetting and masking

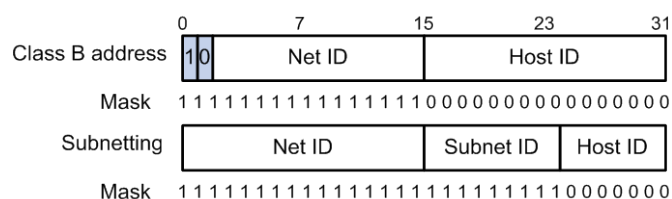
Subnetting divides a network into smaller networks called subnets by using some bits of the host ID to create a subnet ID.

Masking identifies the boundary between the host ID and the combination of net ID and subnet ID.

Each subnet mask comprises 32 bits that correspond to the bits in an IP address. In a subnet mask, consecutive ones represent the net ID and subnet ID, and consecutive zeros represent the host ID.

Before being subnetted, Class A, B, and C networks use these default masks (also called natural masks): 255.0.0.0, 255.255.0.0, and 255.255.255.0, respectively.

Figure 2 Subnetting a Class B network



Subnetting increases the number of addresses that cannot be assigned to hosts. Therefore, using subnets means accommodating fewer hosts.

For example, a Class B network without subnetting can accommodate 1022 more hosts than the same network subnetted into 512 subnets.

- **Without subnetting**—65534 ($2^{16} - 2$) hosts. (The two deducted addresses are the broadcast address, which has an all-one host ID, and the network address, which has an all-zero host ID.)
- **With subnetting**—Using the first nine bits of the host-id for subnetting provides 512 (2^9) subnets. However, only seven bits remain available for the host ID. This allows 126 ($2^7 - 2$) hosts in each subnet, a total of 64512 (512×126) hosts.

IP address assignment

The following are methods available for assigning an IP address to an interface:

- Manual assignment. This chapter describes only manual IP address assignment for interfaces.
- BOOTP. For information about BOOTP, see "Configuring the BOOTP client."
- DHCP. For information about DHCP, see "Configuring the DHCP client."

These methods are mutually exclusive. If you change the IP address assignment method, the new IP address will overwrite the previous address.